

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – EXPERIMENTATION

|                  |                                                                                                                                               |               |                                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | ITA03                                                                                                                                         | Course Title: | Mobile computing                    |
| CO Assessed:     | CO2                                                                                                                                           | Assessment:   | Assessment Tool 2 – EXPERIMENTATION |
| Weightage:       | 30 % of CIA                                                                                                                                   | Total Marks:  | 20 marks                            |
| CO2 Statement:   | Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications. |               |                                     |
| Student Name:    |                                                                                                                                               | Reg. No.:     |                                     |
| Section / Batch: |                                                                                                                                               | Date:         |                                     |

### Code of Conduct

*I Sybil Litzy A (192421110) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: Sybil Litzy A*

### Instructions

- This test contains ONE question.
- Duration: 30 minutes.

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## Annexure A - QUESTIONS

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### 1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunnelling.

Analyse the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

### 2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

### 3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

#### 4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

#### 5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

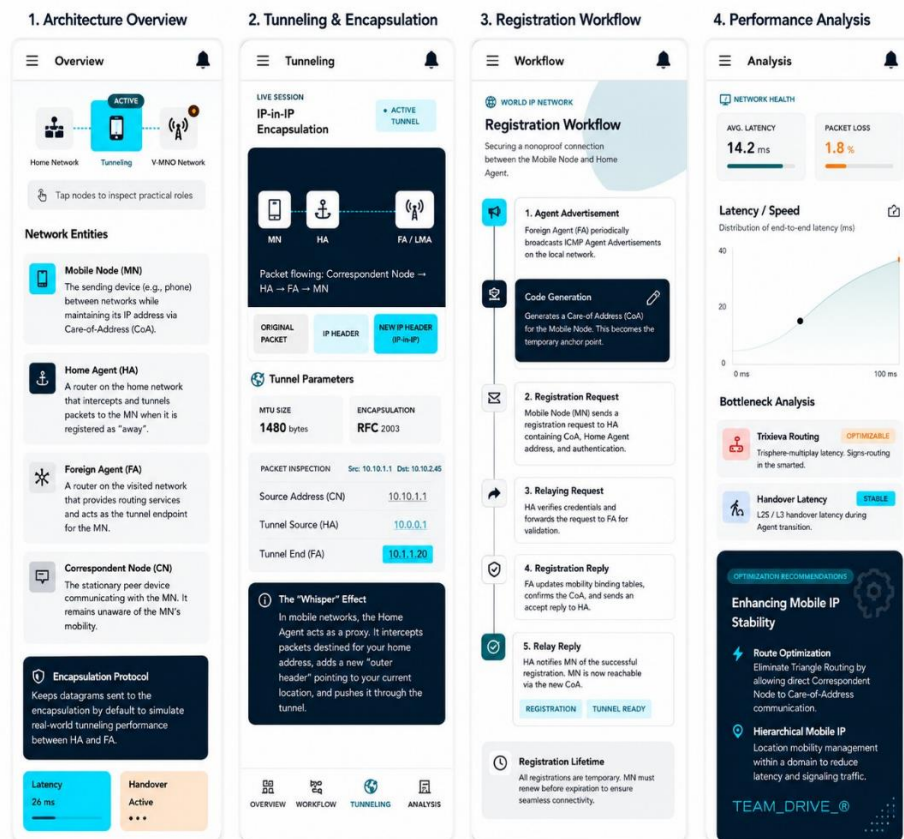
| Criterion                                      | Marks            |
|------------------------------------------------|------------------|
| Design Quality (Figma/Programming Tool)        | 20               |
| Technical Analysis (BL4)                       | 30               |
| Correctness of Mobile Networking Concepts      | 20               |
| GitHub Repository Organization & Documentation | 15               |
| Creativity and Visualization                   | 15               |
| <b>Total</b>                                   | <b>100 Marks</b> |
|                                                |                  |

# ANSWERS:

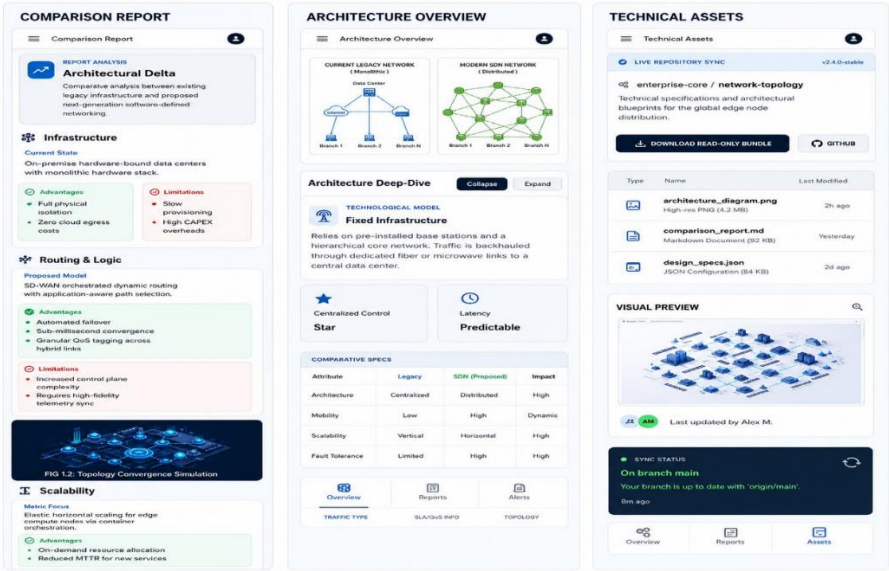
Figma link:

<https://www.figma.com/design/BlV3LdH3V8i00xIguZdutR/Untitled?node-id=0-1&p=f&m=draw>

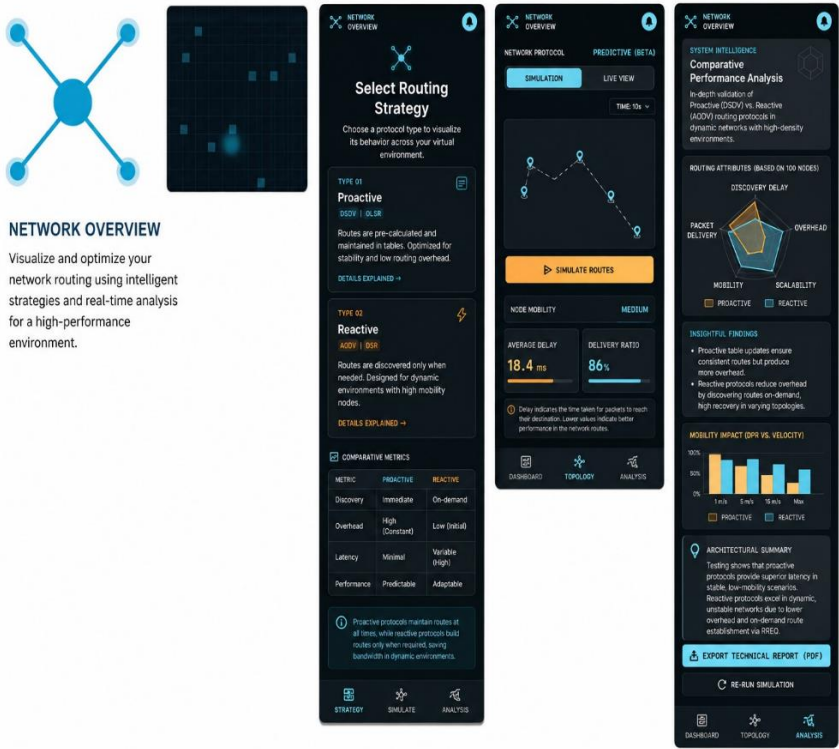
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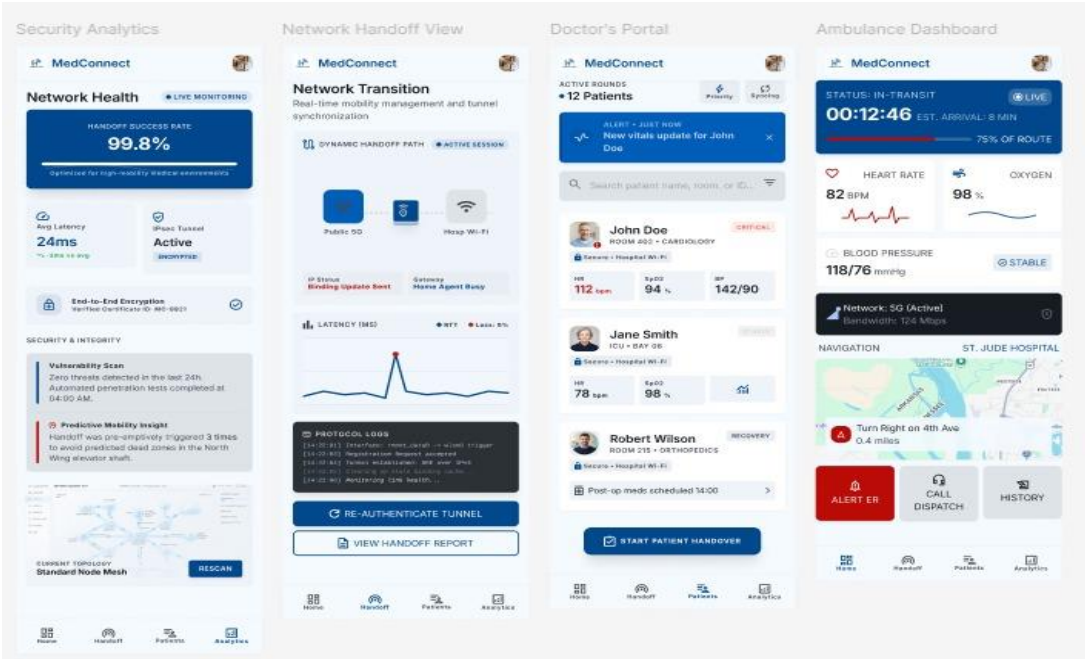
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